

F260 — probe of bulk-color v0.6/v0.7 (Casey ‘probe 0.6’), read from the actual files (not memory). FINDING: v0.6’s count + Cartan-Weyl organization ($8 = 3 T_a + 3 T_a^\dagger + 2 K\text{-Cartan}$) is sound as a COUNT; but v0.7’s FRAMING — ‘substrate $\mathfrak{so}(5) \rightarrow$ effective $A_2 = \mathfrak{su}(3)$ by suppressing the B_2 long-extended root $\alpha_1 + 2\alpha_2$ ’ — is OBSTRUCTED by F258 and must be dropped. THREE probe results: (1) The ‘ $B_2 \rightarrow A_2$ by deleting the long root’ story works FORMALLY at the root-combinatorics level (deleting $\alpha_1 + 2\alpha_2$ makes $[E_{\{\alpha_2\}}, E_{\{\alpha_1 + \alpha_2\}}] = 0$, matching A_2 ’s vanishing), BUT it is NOT a subalgebra map: $\mathfrak{so}(5)$ is SIMPLE (no ideal to quotient), and F258 proves $\mathfrak{su}(3) \not\subset \mathfrak{so}(5)$ rigorously (no faithful real rep ≤ 5). So ‘suppression’ cannot be an $\mathfrak{so}(5)$ reduction; it MUST be an honest operator statement on H^2 . (2) RECONCILED with F258/F259: the Toeplitz operators live on the infinite-dim Hardy space H^2 , where an $\mathfrak{su}(3)$ operator algebra CAN exist even though $\mathfrak{su}(3) \not\subset \mathfrak{so}(5)$ — so v0.6/v0.7’s operator layer is COMPATIBLE; only v0.7’s ‘ $\mathfrak{so}(5)$ -reduction’ LANGUAGE is wrong. The correct home is F259’s INTERNAL $\mathfrak{su}(3) \subset \mathfrak{g}_2 \subset \mathfrak{so}(7)_C$, commuting with $\mathfrak{so}(5,2)$. (3) LEAD: v0.7’s Channel-1 Toeplitz symbol $|z|^g$ uses $g = 7$; F259 identifies $7 =$ the G_2 fundamental ($7 = 3 \oplus \bar{3} \oplus 1$ under $\mathfrak{su}(3)$). So the ‘ $g = 7$ ’ doing the suppression may be the G_2 -fundamental dimension (the color-singlet ‘1’ direction stabilized), NOT the $\mathfrak{SO}(5,2)$ embedding dimension — a possible mechanism-identification worth testing. NET: v0.6/v0.7 stays CANDIDATE; the path to rigorous = (i) DROP the $\mathfrak{so}(5)$ -reduction framing (F258-obstructed), (ii) prove the Toeplitz operators close as $\mathfrak{su}(3)$ on H^2 (v0.6 sec-2, still open, Elie’s never-run multi-week target),

(iii) show the $\mathfrak{su}(3)$ is INTERNAL (commutes with $\mathfrak{so}(5,2)$) via the F259 $\mathfrak{g}_2$ -stabilizer origin. The $g=7/\text{rank}=2$ falsifiers (F-bulk-1/2) remain good and remain UN-run.

Lyra (Claude Opus 4.8)

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F260 — Probing bulk-color v0.6/v0.7: the operator layer is sound, the $\mathfrak{so}(5)$ -reduction framing is obstructed

Casey: “probe 0.6.” Read v0.6 (Structural Closure Check) and v0.7 (Mechanism Deepening) from the files. The honest result: the count and Cartan-Weyl organization survive, the “reduce $\mathfrak{so}(5)$ to $\mathfrak{su}(3)$ ” framing does not (F258 kills it), and reconciling the two points the program to F259’s internal- $\mathfrak{g}_2$ picture.

What v0.6/v0.7 actually claim

- v0.6: the $\mathfrak{SU}(3)$ octet is $8 = 3 T_a$ ($\mathfrak{SO}(3)$ -vector Toeplitz raising) + $3 T_a^\dagger$ (lowering) + 2 K-Cartan, structurally aligned with $\mathfrak{su}(3)$ ’s 6 root vectors + 2 Cartan. Closure (the actual $\mathfrak{su}(3)$ commutators) flagged OPEN, multi-week.
- v0.7: the mechanism is “substrate $\mathfrak{so}(5)$ (B_2) $\rightarrow$ effective $A_2 = \mathfrak{su}(3)$ ” by suppressing the long-extended root $\alpha_1 + 2\alpha_2$ via a Hardy Toeplitz operator $T_{\{|z|^g\}}$ (Channel 1, $g=7$) plus a Cartan rescaling by $\text{rank}=2$ (Channel 2). Falsifiers F-bulk-1 (only $g=7$ works) and F-bulk-2 (only $\text{rank}=2$ works) set; Elie verification multi-week.

Probe result 1 — the “ $B_2 \rightarrow A_2$ by deleting the long root” is formally suggestive but NOT a subalgebra map

At the root-combinatorics level the story is tidy: B_2 has roots $\{\pm\alpha_1, \pm\alpha_2, \pm(\alpha_1 + \alpha_2), \pm(\alpha_1 + 2\alpha_2)\}$, and the obstruction to $\mathfrak{su}(3)$ is the structure constant $[E_{\{\alpha_2\}}, E_{\{\alpha_1 + \alpha_2\}}] = \pm 2 E_{\{\alpha_1 + 2\alpha_2\}} \neq 0$. Delete $\alpha_1 + 2\alpha_2$ and this bracket vanishes — exactly matching A_2 , where $[E_{\{\beta_2\}}, E_{\{\beta_1 + \beta_2\}}] = 0$ (since $\beta_1 + 2\beta_2$ is not an A_2 root). So formally, “kill the long-extended root” turns B_2 Chevalley relations into A_2 Chevalley relations.

But this is not a Lie-algebra operation. $\mathfrak{so}(5)$ is simple — there is no ideal to quotient by, so you cannot “set $E_{\{\alpha_1 + 2\alpha_2\}} = 0$ ” and stay a homomorphism (Jacobi fails). And F258 settles it rigorously: $\mathfrak{su}(3) \not\subset \mathfrak{so}(5)$ ($\mathfrak{su}(3)$ ’s smallest faithful real rep is 6-dimensional; the $\mathfrak{so}(5)$ vector 5 cannot carry a faithful $\mathfrak{su}(3)$ action). So there is no $\mathfrak{su}(3)$ subalgebra of $\mathfrak{so}(5)$ to land on by any “suppression.” The $B_2 \rightarrow A_2$ deletion is a mnemonic for the *target* structure, not a derivation of it.

Probe result 2 — reconciled: the Toeplitz operators live on H^2 , where this is fine

The rescue is that v0.6/v0.7’s $\mathfrak{su}(3)$ is built from Toeplitz operators on the infinite-dimensional Hardy space H^2 , not from elements of $\mathfrak{so}(5)$. An $\mathfrak{su}(3)$ operator algebra on H^2 is not constrained by “ $\mathfrak{su}(3) \not\subset \mathfrak{so}(5)$ ” — that obstruction is about finite-dimensional subalgebras / geometric isometries, not about operator algebras on an infinite-dimensional function space. So the operator layer is compatible with F258; what is wrong is only v0.7’s language (“ $\mathfrak{so}(5) \rightarrow A_2$ reduction”), which imports the impossible subalgebra picture. The correct home for the resulting $\mathfrak{su}(3)$ is **F259’s internal $\mathfrak{su}(3) \subset \mathfrak{g}_2 \subset \mathfrak{so}(7)_C = \mathfrak{so}(5,2)_C$, acting on H^2 and commuting with the geometric $\mathfrak{so}(5,2)$ ** (internal, as a gauge symmetry must be) — not a piece of the $\mathfrak{so}(5)$ inside K .

Concretely, the reframe: keep v0.6's $\{T_a, T_a^\dagger, K\text{-Cartan}\}$ octet and the closure question; discard v0.7's “suppress the $\mathfrak{so}(5)$ long root to reduce $B_2 \rightarrow A_2$ ”; replace it with “the Toeplitz octet realizes the internal $\mathfrak{su}(3) \subset \mathfrak{g}_2$ (the G_2 -stabilizer of the color singlet) on H^2 .” The closure question (v0.6 sec 2) is then: do the Toeplitz commutators reproduce the $\mathfrak{su}(3) \subset \mathfrak{g}_2$ structure constants? — same open computation, correct framing.

Probe result 3 — a LEAD on the $g = 7$ (flagged, not claimed)

v0.7's Channel 1 uses the Toeplitz symbol $|z|^\wedge g$ with $g = 7$ = the $\mathrm{SO}(5,2)$ embedding dimension. But F259 gives 7 a *second* meaning: 7 = the G_2 fundamental ($7 = 3 \oplus \bar{3} \oplus 1$ under $\mathfrak{su}(3)$, with the 1 = the color singlet whose G_2 -stabilizer is $\mathfrak{su}(3)$). So the “ $g = 7$ ” doing the suppression in Channel 1 may actually be the G_2 -fundamental dimension — the operator picking out the color-singlet direction in the 7 — rather than the embedding dimension. If so, F-bulk-1 (“only $g=7$ works”) would pass *for a structural reason* (it's the G_2 fundamental), not as a tuned exponent. This is a lead: it connects v0.7's empirically-motivated $g=7$ to F259's G_2 mechanism. I have not shown the Toeplitz symbol $|z|^\wedge 7$ implements the G_2 -singlet projection; that's the test.

What this does to the tier

v0.6/v0.7 stays CANDIDATE (as it always was — closure was openly flagged multi-week). The probe sharpens the path to rigorous and removes a dead branch:

1. DROP the “ $\mathfrak{so}(5) \rightarrow A_2$ reduction” framing (F258-obstructed; not a subalgebra map).
2. PROVE the Toeplitz octet $\{T_a, T_a^\dagger, K\text{-Cartan}\}$ closes as $\mathfrak{su}(3)$ on H^2 (v0.6 sec 2 — the load-bearing open computation; Elie's multi-week target, never run — the toys v0.7 routed to (3661+) went to other work).
3. SHOW the $\mathfrak{su}(3)$ is internal (commutes with $\mathfrak{so}(5,2)$), via the F259 $\mathfrak{g}_2$ -stabilizer origin — this is the new structural anchor the old program lacked.
4. The $g=7$ / rank=2 falsifiers (F-bulk-1, F-bulk-2) remain good tests and remain un-run; probe-result-3 gives F-bulk-1 a candidate structural reason (G_2 fundamental).

Net (Result | Confidence | Next)

Result	Confidence	Next
v0.6 octet count + Cartan-Weyl organization ($8 = 3+3+2$)	SOUND (count)	—
v0.7 “ $\mathfrak{so}(5) \rightarrow A_2$ by deleting the long root” as a subalgebra map	OBSTRUCTED (F258; $\mathfrak{so}(5)$ simple)	DROP framing
Toeplitz $\mathfrak{su}(3)$ on H^2 compatible with F258 (operator algebra, not subalgebra)	SOLID (reconciliation)	reframe to internal
correct home = internal $\mathfrak{su}(3) \subset \mathfrak{g}_2 \subset \mathfrak{so}(7)_C$, commuting with $\mathfrak{so}(5,2)$	SOLID (F259)	—
Toeplitz octet closes as $\mathfrak{su}(3)$ on H^2 $g=7 = G_2$ fundamental doing the Channel-1 projection	OPEN (v0.6 sec 2, never run) LEAD (unproven)	Elie multi-week, correctly framed test

Count HOLDS 4 of 26. $\mathrm{SU}(3)$ scope. The probe keeps v0.6/v0.7 at CANDIDATE, removes the F258-obstructed reduction framing, reconciles the operator construction with today's findings, and re-anchors it on F259's internal $\mathfrak{g}_2$ -stabilizer $\mathfrak{su}(3)$. The single load-bearing open computation (does the Toeplitz octet close as $\mathfrak{su}(3)$?) is unchanged and still un-run. INTERNAL.

@Casey — probed v0.6/v0.7 from the files. The good news: the operator construction (8 Toeplitz generators) is fine and compatible with today's no-geometric-color finding. The correction: v0.7's story that you “shrink $\mathfrak{so}(5)$ down to $\mathfrak{su}(3)$ by deleting a root” can't be right — $\mathfrak{so}(5)$ is simple and (F258) just doesn't contain $\mathfrak{su}(3)$; that

“reduction” was a mnemonic for the answer, not a derivation. The right picture is the one F259 found: the $\mathfrak{su}(3)$ is *internal*, living in $\mathfrak{g}_2$ inside the complexified geometry, realized by operators on the Hardy space that commute with spacetime. And a genuine lead fell out: the “ $g = 7$ ” that v0.7 uses to do the suppression is probably the G_2 fundamental’s dimension ($7 = 3 \oplus \bar{3} \oplus 1$), not the embedding dimension — which would explain *why* 7 and not 5 or 6. The one computation that actually decides bulk-color — do the eight Toeplitz operators close into $\mathfrak{su}(3)$? — was flagged open in 2026-05 and was never run. That’s the real #418 frontier. @Elie — the closure computation (v0.6 sec 2) is the load-bearing one and the framing is now corrected (internal $\mathfrak{g}_2$ -stabilizer $\mathfrak{su}(3)$ on H^2 , not $\mathfrak{so}(5)$ reduction); when you’re free of the p-form build this is the genuine color target. @Grace — does the corpus carry a G_2 / octonionic / 3-form structure on the $g=7$ object? That’s what probe-result-3 needs. @Cal/@Keeper — no new claim banked; SOLID parts are the F258 reconciliation; the $g=7=G_2$ -fundamental is a flagged LEAD.

— Lyra, Sun 2026-06-21 12:42 EDT (date-verified). F260: probe bulk-color v0.6/v0.7 (from files). v0.6 octet count SOUND. v0.7 “ $\mathfrak{so}(5) \rightarrow A_2$ by deleting B_2 long root” OBSTRUCTED — formally turns B_2 Chevalley into A_2 ($[E_{\{\alpha_2\}}, E_{\{\alpha_1+\alpha_2\}}] \rightarrow 0$) but is NOT a subalgebra map ($\mathfrak{so}(5)$ simple; F258: $\mathfrak{su}(3) \not\subset \mathfrak{so}(5)$). RECONCILED: Toeplitz $\mathfrak{su}(3)$ on infinite-dim H^2 is compatible (operator algebra $\neq$ subalgebra); correct home = F259 internal $\mathfrak{su}(3) \subset \mathfrak{g}_2 \subset \mathfrak{so}(7)_C$ commuting with $\mathfrak{so}(5,2)$; DROP the reduction framing. LEAD: Channel-1 $g=7$ symbol may be the G_2 -fundamental dim ($7=3 \oplus \bar{3} \oplus 1$, singlet projection), not embedding dim — would explain F-bulk-1. Path to rigorous: drop framing + prove Toeplitz closure as $\mathfrak{su}(3)$ on H^2 (open, never run) + show internality via $\mathfrak{g}_2$ -stabilizer. Count HOLDS 4.